C1 - The pattern audit, at scale

Project proposal | Week 9
Members: Saida Musaeva, Mikhail Tereshchenko, Anna Tikhonova
Research question: Do training only relation-pattern measurements predict per-relation KGE test MRR for relation types held out from the performance-prediction model, beyond training frequency?

Task and success criterion

Measure symmetry, inversion, composition, and 1-to-N structure for each relation, then test whether these features predict the performance of TransE, DistMult, ComplEx, and RotatE. The primary experiment is on FB15k-237. Define MAE improvement as baseline MAE minus pattern-model MAE (positive favors patterns). Claim improvement only if its paired bootstrap 95% confidence interval over held-out relations is above zero; otherwise report the result as inconclusive. R-squared is secondary.

Data and method

Use the standard PyG train/validation/test splits of FB15k_237 and WordNet18RR. Compute all features from training triples only, using set-based counts: **symmetry** = fraction of (h,r,t) triples whose reverse (t,r,h) is in training; **inversion** = maximum, over s != r, of the fraction of r-pairs (h,t) for which (t,s,h) is in training; **composition** = for target relation r, the maximum confidence over ordered body relations (p,q): a body supports (h,t) if training contains (h,p,z) and (z,q,t) for some z; confidence is the fraction of distinct body-supported pairs also present under r. Consider only bodies with at least 10 distinct (h,t) pairs. **1-to-N** = mean number of distinct tails per head. Freeze these definitions before evaluation. Retain relations with at least 50 training triples and report exclusions.
Train TransE, DistMult, ComplEx, and RotatE with the same predeclared tuning budget per model; select hyperparameters on validation filtered MRR. Run five fixed seeds (0-4). For each relation, compute standard filtered test MRR over head- and tail-prediction queries and record the number of test queries. Use mean MRR across seeds as the primary target and report between-seed variation separately.

Evaluation plan

For each KGE model on FB15k_237, compare (a) Ridge regression on log training-triple count and (b) Ridge using that count plus the four pattern features. Use five-fold cross-validation split by relation, standardize predictors and tune regularization within each training fold, and use identical held-out relations for both models. Report out-of-fold MAE, R-squared, the confidence interval for MAE improvement, and ablations removing one pattern feature at a time. Test MRR is used only as the outcome, not to select KGE models or features.
WN18RR has only 11 relation types, so treat it as an exploratory replication: report per-relation feature values and MRRs, and describe leave-one-relation-out results without claiming reliable meta-prediction.

Novelty, risk, and reproducibility

Prior work compares KGE performance across relational-pattern subsets on these benchmarks [1]. Our contribution is a leakage-controlled, out-of-sample test of whether graded, training-only relation features predict test MRR beyond log relation frequency, with model-specific comparisons and feature ablations; we do not claim that the general pattern-performance link is new. **Risk:** sparse evidence may make feature estimates and per-relation MRR noisy. **Fallback:** report support counts, apply the 50-triple threshold, and treat composition as exploratory if unstable; retain the frequency-only baseline. A null result is informative if measured honestly.

Reference

[1] Jin et al. (2023), *A Comprehensive Study on Knowledge Graph Embedding over Relational Patterns Based on Rule Learning*. https://arxiv.org/abs/2308.07889